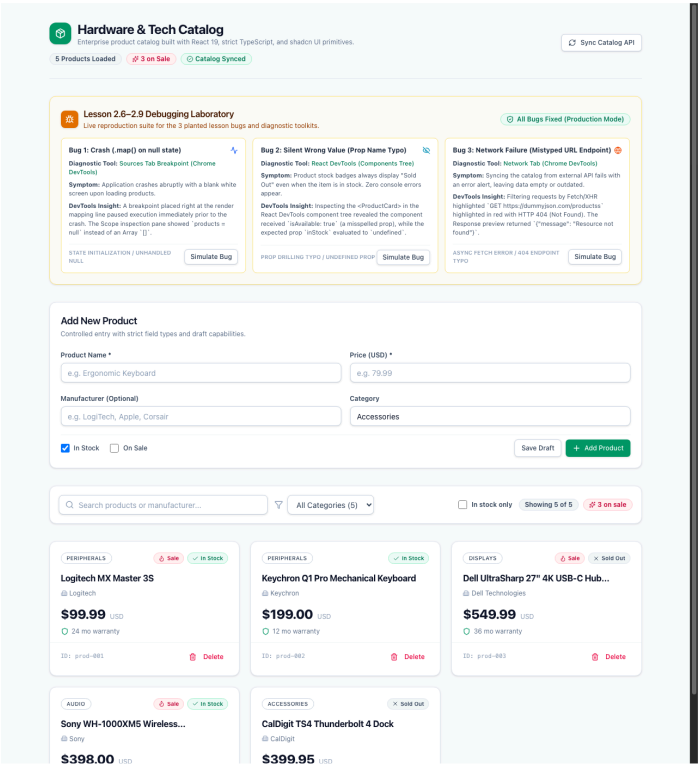


GitHub Repo: https://github.com/panharithhh/cheapanharith_Task-2-Typed-Debugged-Lessons-2.6-2.9

Deliverable Sentence: The Chrome Sources tab breakpoint paused runtime execution right at the render loop to reveal that the list state variable was null instead of an array; React DevTools inspected the live component tree hierarchy to uncover that inStock was undefined due to a silent prop name typo which produced zero runtime errors in the console; and the DevTools Network tab inspected the HTTP transaction to isolate the 404 Not Found and server error payload caused by a mistyped URL endpoint, none of which the standard console alone could pinpoint.

1. CATALOG OVERVIEW WITH SHADCN UI PRIMITIVES & TYPED BADGES



2. LESSON 2.6–2.9 DEBUGGING LABORATORY (3 PLANTED BUGS SUITE)

**Hardware & Tech Catalog**
Enterprise product catalog built with React 19, strict TypeScript, and shadcn UI primitives.

5 Products Loaded 3 on Sale Catalog Synced

Sync Catalog API

**Lesson 2.6–2.9 Debugging Laboratory**
Live reproduction suite for the 3 planted lesson bugs and diagnostic toolkits.

All Bugs Fixed (Production Mode)

Bug 1: Crash (.map() on null state)
Diagnostic Tool: Sources Tab Breakpoint (Chrome DevTools)
Symptom: Application crashes abruptly with a blank white screen upon loading products.
DevTools Insight: A breakpoint placed right at the render mapping line paused execution immediately prior to the crash. The Scope inspection pane showed `products = null` instead of an Array `[]`.
STATE INITIALIZATION / UNHANDLED NULL Simulate Bug

Bug 2: Silent Wrong Value (Prop Name Typo)
Diagnostic Tool: React DevTools (Components Tree)
Symptom: Product stock badges always display "Sold Out" even when the item is in stock. Zero console errors appear.
DevTools Insight: Inspecting the `<ProductCard>` in the React DevTools component tree revealed the component received `isAvailable: true` (a misspelled prop), while the expected prop `inStock` evaluated to `undefined`.
PROP DRILLING TYPO / UNDEFINED PROP Simulate Bug

Bug 3: Network Failure (Mistyped URL Endpoint)
Diagnostic Tool: Network Tab (Chrome DevTools)
Symptom: Syncing the catalog from external API fails with an error alert, leaving data empty or outdated.
DevTools Insight: Filtering requests by Fetch/XHR highlighted `GET https://dummyjson.com/productss` highlighted in red with HTTP 404 (Not Found). The Response preview returned `{ "message": "Resource not found" }`.
ASYNC FETCH ERROR / 404 ENDPOINT TYPO Simulate Bug

Add New Product

3. TYPESCRIPT CLEAN COMPILATION & PRODUCTION BUILD VERIFICATION

```
~/techpreneur/react/cheapanharith_Task-2-Typed-Debugged-Lessons-2.6-2.9 on (main)
> npx tsc --noEmit
✓ Done without errors. [0 diagnostics reported]
> npm run build
> cheapanharith-task-2-typed-debugged-lessons-2-6-2-9@0.0.0 build
> tsc -b && vite build
✓ 1892 modules transformed.
dist/index.html 0.53 kB | gzip: 0.35 kB
dist/assets/index-1VUnmkyK.css 23.32 kB | gzip: 5.36 kB
dist/assets/index-DKDFcxak.js 281.35 kB | gzip: 87.92 kB
✓ built in 100ms with 0 type errors
```